

Password Policy Enforcement

Cybersecurity Portfolio Project

1. Project Overview

This project focused on enforcing Windows password security policies using Local Group Policy Editor and Local Users and Groups management tools. The project demonstrates how password complexity and minimum password length policies help strengthen account security and reduce weak password usage.

2. Objective

The goal of this project was to configure and test Windows password policy enforcement settings by enabling password complexity requirements and minimum password length restrictions.

3. Tools Used

- Windows 10
- Local Group Policy Editor ([gpedit.msc](#))
- Computer Management
- VirtualBox

4. Skills Demonstrated

- Password Policy Enforcement
- Windows Security Administration
- Identity & Access Management (IAM)
- Security Hardening
- Endpoint Security
- Account Security Controls

5. Lab Environment

The lab environment was created using a Windows 10 virtual machine inside VirtualBox. Administrative tools such as Local Group Policy Editor and Local Users and Groups were used to configure and test password policies.

6. Steps Performed

1. Opened the Run dialog box.
2. Launched Local Group Policy Editor using `gpedit.msc`.
3. Navigated to:
 - Computer Configuration
 - Windows Settings
 - Security Settings
 - Account Policies
 - Password Policy
4. Enabled “Password must meet complexity requirements.”
5. Configured a minimum password length of 8 characters.
6. Attempted to create a user account with a weak password.
7. Verified that Windows blocked the weak password based on the enforced policy.

7. Screenshots

Figure 1 — Creating a New User Account

- Opened Local Users and Groups.
- Attempted to create a user named “mary”.

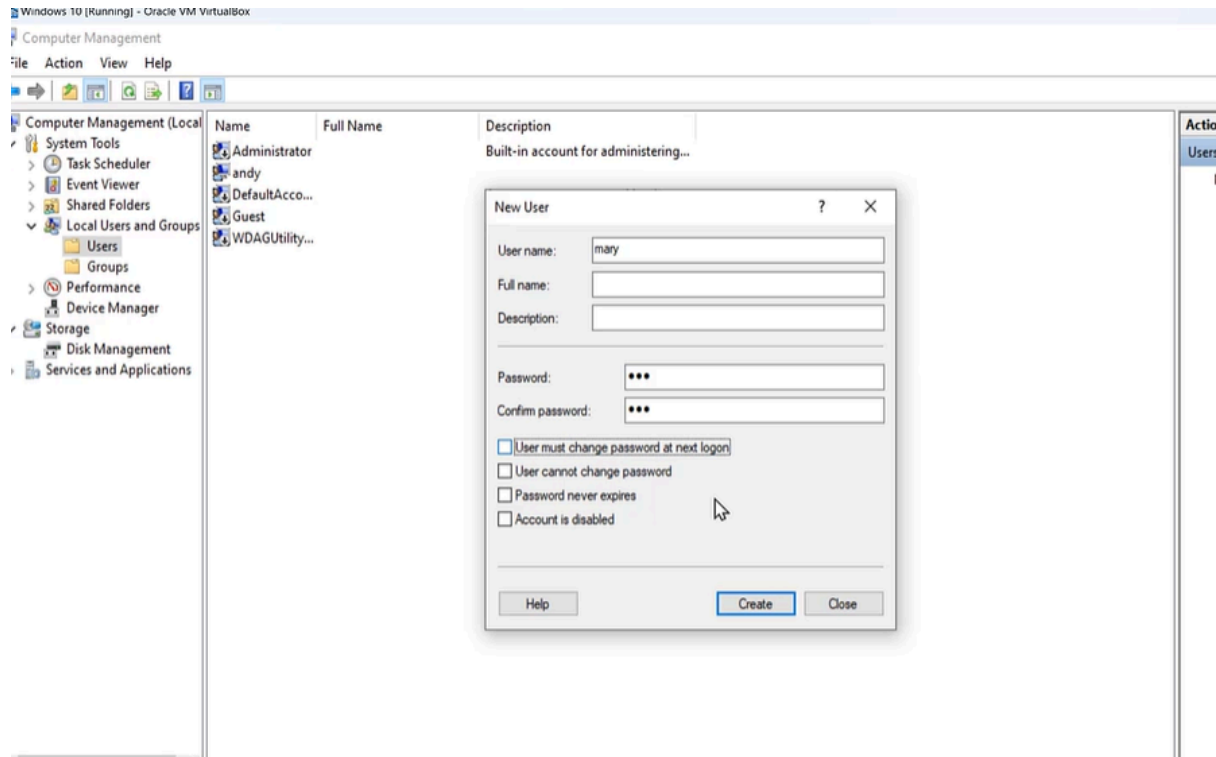


Figure 2 — Opening Group Policy Editor

- Used `gpedit.msc` to access Local Group Policy Editor.

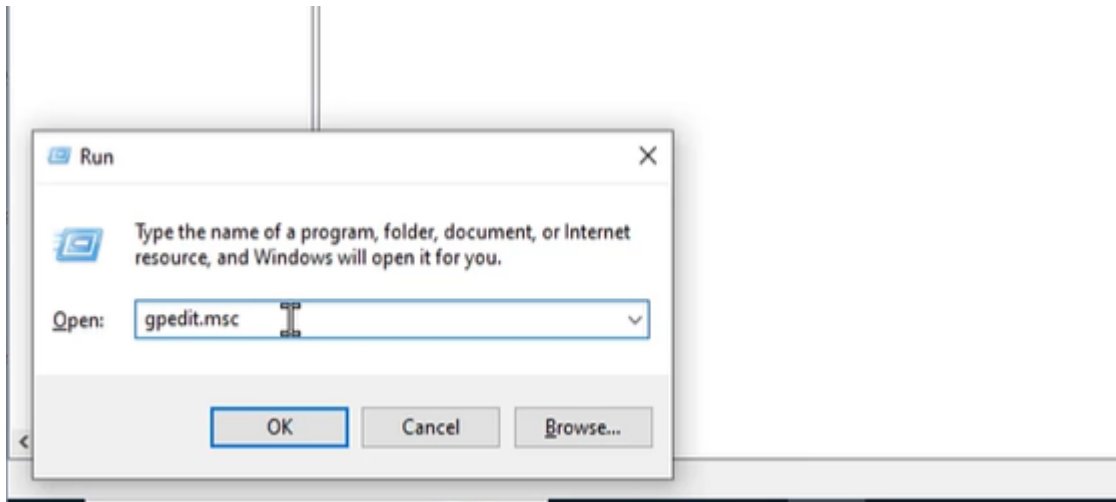


Figure 3 — Password Complexity Requirements Enabled

- Navigated to Password Policy settings.
- Enabled password complexity requirements.

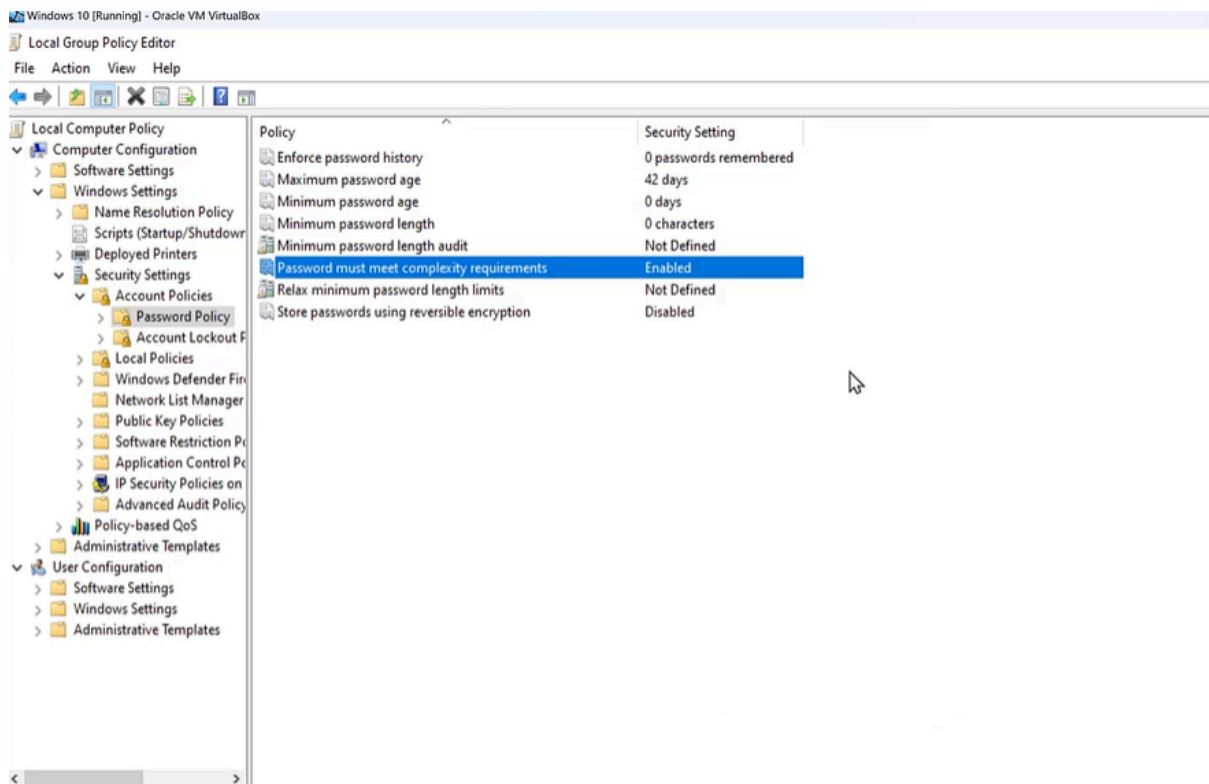


Figure 4 — Configuring Minimum Password Length

- Set minimum password length to 8 characters.

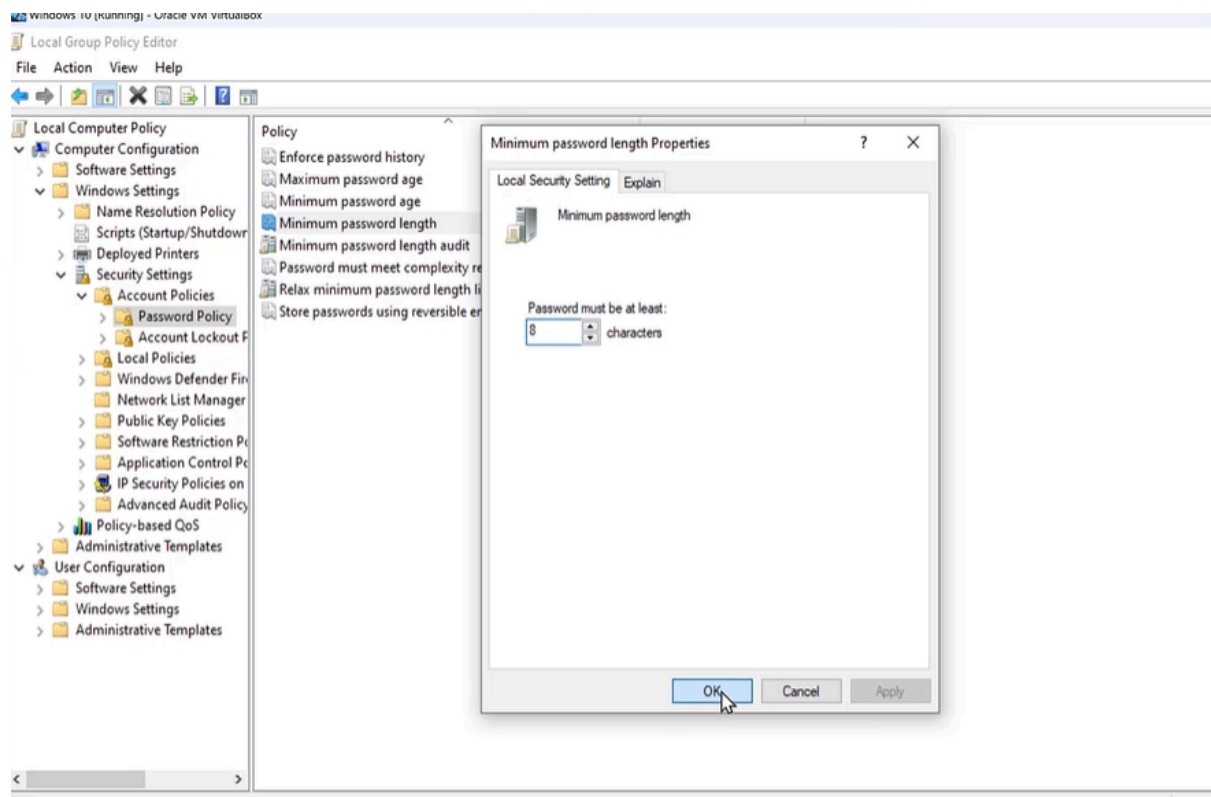
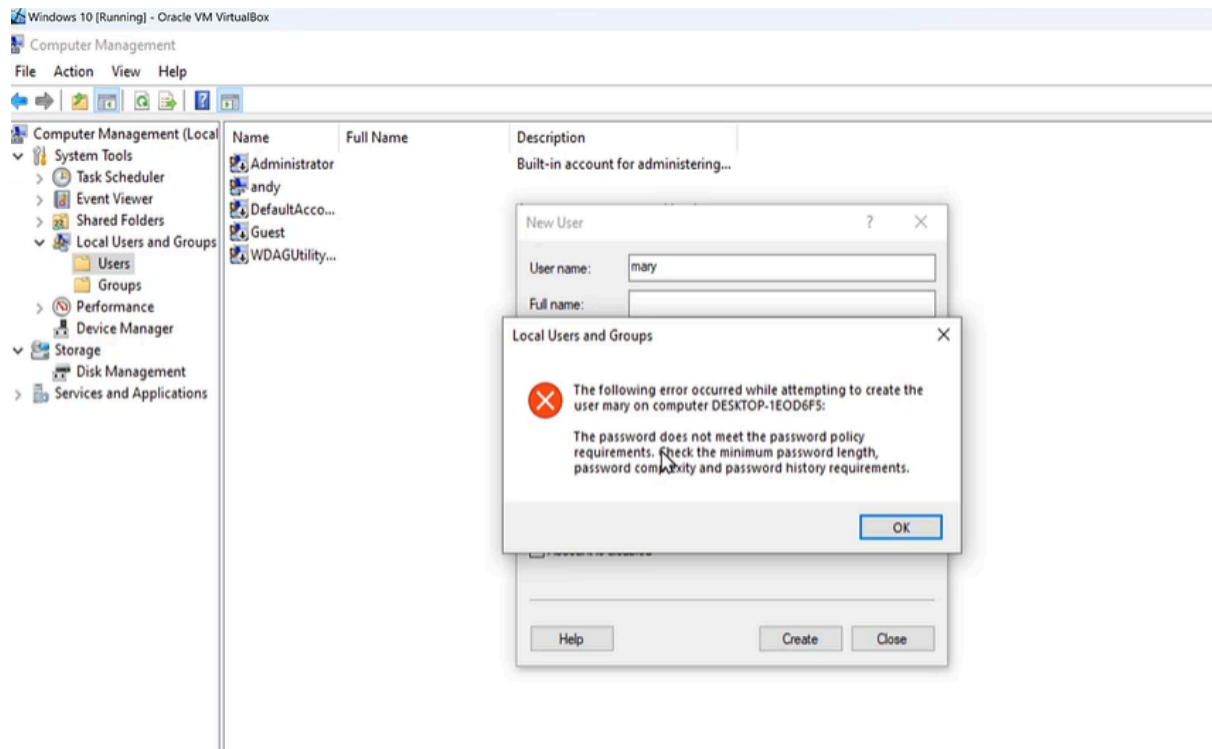


Figure 5 — Password Policy Enforcement

- Windows blocked creation of a weak password.
- Error confirmed that password requirements were enforced successfully.



8. Security Benefits

Password policies improve account security by reducing weak passwords and enforcing stronger authentication standards.

These controls help:

- Prevent unauthorized access
- Reduce brute-force risks
- Improve compliance
- Strengthen identity security

9. What I Learned

I learned how Windows security policies can enforce password standards and prevent users from creating weak passwords.

I also gained experience configuring account security settings using Local Group Policy Editor.

10. Conclusion

This project demonstrated practical cybersecurity skills related to password security enforcement, Windows administration, and identity and access management controls using Local Group Policy settings.